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Project Proposal

AirNova

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1. What?

AirNova is a machine learning project designed to predict air pollution levels—specifically PM2.5 concentration—in Sofia, Bulgaria. The project uses publicly available air quality data and combines it with weather variables such as temperature, humidity, wind speed, wind direction, atmospheric pressure, and rainfall. By learning from historical patterns, the resulting model will predict PM2.5 levels and help identify environmental or weather conditions most associated with poor air quality.

2. Why?

AirNova's project is relevant personally because Sofia, the capital of my home country, frequently struggles with air quality issues, particularly during winter and periods of heavy traffic. Like many residents, I have many family members and friends who live there, and it often causes health problems such as respiratory issues and allergies. Poor air quality also affects daily activities and overall well-being, making it difficult to enjoy outdoor spaces or engage in physical exercise safely. This project aims to provide better understanding and prediction of pollution levels, which can help residents take timely precautions and improve their quality of life. By using open data and machine learning, AirNova addresses a real and urgent local concern while applying the skills and knowledge I am gaining through my studies.

3. Who?

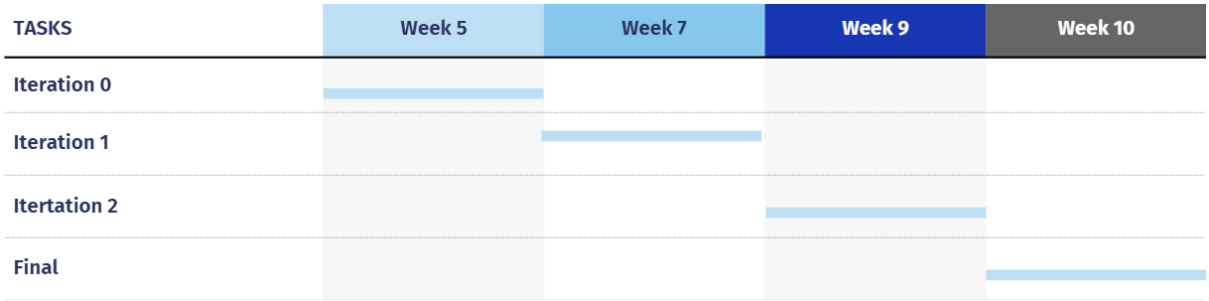
- **Primary Stakeholder:** Anna Ivanova

Anna Ivanova is a 22-year-old computer science student living in Sofia who suffers from asthma. Balancing her studies and daily life, she experiences worsened asthma symptoms during periods of poor air quality. Anna wants accurate predictions so she can avoid outdoor exposure when pollution is high and manage her health effectively while maintaining her busy schedule. AirNova will provide her with timely information to make smarter decisions about his environment and well-being.

- **General public**

The public in Sofia and surrounding areas who are exposed to fluctuating air quality levels will also benefit from AirNova. This includes families with young children, elderly individuals, people with respiratory or cardiovascular conditions, outdoor workers, and anyone interested in protecting themselves from harmful pollution.

4. When?



- **Iteration 0:** Define project scope, draft initial proposal, gather datasets, and set up the development environment.
- **Iteration 1:** Apply feedback, clean and preprocess data, implement initial machine learning models, and conduct exploratory analysis
- **Iteration 2:** Refine models, improve, evaluate results, and generate visualizations.
- **Iteration 3:** Finalize model, polish results and presentation, and prepare demo focused on user-friendly insights.

5. How?

Data will be collected from publicly available sources for air quality measurements (PM2.5, PM10) in Sofia and historical weather data. The datasets will include relevant features like pollutant concentrations, temperature, humidity, wind speed and direction, atmospheric pressure, and precipitation. Data cleaning will address missing values, correct inconsistent formats, and align timestamps to create a unified, hourly time series. Feature engineering will create meaningful variables such as lagged pollution levels, time of day, and categorical wind directions. Machine learning models such as Linear Regression and Decision Trees will be developed using a train-test split of the data. The final results will include visualizations and analyses highlighting key factors that influence air pollution, helping generate actionable insights for residents and city planners.

6. Domain understanding

6.1 Research Question

Which air pollutant in Sofia shows the strongest relationship with meteorological and environmental factors, and how can this understanding guide predictive modeling?

Sub-questions

- Among key pollutants (PM2.5, PM10, NO_2, O_3, CO, SO_2), which shows the strongest correlation with meteorological variables such as temperature, humidity, wind speed, and pressure?
- How do seasonal and temporal patterns (e.g., winter vs. summer, day vs. night) influence PM2.5 and other pollutant concentrations in Sofia?
- Which meteorological factors most significantly contribute to high PM2.5 concentrations, and how can these insights improve air pollution forecasting for Sofia?

6.2 Exploratory Research:

Before modelling, I will explore the air quality and weather data to understand its size, completeness, and structure. I will check for missing or incorrect values and analyze the distribution of key variables like PM2.5, temperature, humidity, and wind speed. Correlations between pollutants and weather features will be examined to identify influential factors. Time-series plots will reveal patterns and trends over days or seasons. This exploration will guide feature selection, data cleaning, and transformations needed for accurate modelling.

6.3 Resources

- Datasets:
  - ✓ **OpenWeatherMap Air Pollution API:** Air quality data for Sofia, including PM2.5 and PM10 measurements.

Link: https://open-meteo.com/en/docs/air-quality-api

  - ✓ **Weather data:** Historical weather records for Sofia from **Open-Meteo Weather API**, containing temperature, humidity, wind speed/direction, pressure, and precipitation.

Link: https://open-meteo.com/en/docs/historical-weather-api

- Tools & Libraries:

- ✓ Python
- ✓ Jupyter Notebook
- ✓ Libraries: Pandas (data handling)

- AI & Support:
  - ✓ ChatGPT for coding assistance and troubleshooting

7. Analytic approach

7.1 Target Variable

The main target variable for AirNova is the hourly PM2.5 concentration one hour ahead **(PM2.5_{t+1})**.

This short-term forecast provides timely information for immediate decisions, such as avoiding outdoor exposure during high pollution periods.

7.2 Type of Problem

This is a regression problem, as the goal is to predict a continuous numeric value ($\mu g/m^3$).

7.3 Machine Learning Models:

To ensure robust results, multiple models will be trained and compared:

Linear Regression:

A simple and interpretable baseline model that shows the direct relationship between weather variables and PM2.5 levels.

Decision Trees:

Capture nonlinear relationships and interactions between variables.

Provide feature importance, which will help identify which weather conditions most strongly influence PM2.5.

Random Forests:

Combine many decision trees to improve accuracy and reduce overfitting.

Expected to give the most reliable predictions and highlight the most important features.

7.4 Expected Outcomes After Training:

- **Performance Metrics:**

R^2 scores to measure how accurate each model is.

- **Feature Importance:**

Identification of the top weather factors affecting PM2.5 levels, shown as coefficients (for Linear Regression) or importance scores (for tree-based models).

- **Visual Comparisons:**

Graphs comparing predicted vs. actual PM2.5 values to show how well the models follow real trends.

8. Data Requirements

8.1 Objectives

To predict PM2.5 pollution levels in Sofia using weather and air quality features to improve understanding and help inform public health decisions.

8.2 Factors and Features

Air Quality Pollutants: Concentrations of PM2.5, PM10, NO2, O3, CO, and SO2 measured at various monitoring stations. These pollutants are central to understanding air quality and its impact on health.

Meteorological Variables: Temperature, humidity, wind speed, wind direction, atmospheric pressure, and precipitation. These factors influence the dispersion, transformation, and concentration of pollutants in the air.

Temporal Features: Timestamp data capturing the date and time of measurements, allowing models to learn daily and seasonal pollution patterns. Features such as hour of day, day of week, and seasonality will be derived to enhance predictions.

Location Information: Name and geographic coordinates (latitude and longitude) of monitoring stations to consider spatial variability in pollution levels.

time	# pm2_5	# pm10	# carbon_monoxide	# nitrogen_dioxide
2023-09-25 00:00:00	15.4	29.7	242.0	8.4
2023-09-25 01:00:00	13.8	28.7	251.0	8.1
2023-09-25 02:00:00	12.7	27.3	269.0	6.0
2023-09-25 03:00:00	8.8	28.3	270.0	5.9
2023-09-25 04:00:00	10.9	24.3	244.0	4.6

# sulphur_dioxide	# ozone	# temperature_2m	# relative_humidity_2m	# windspeed_10m
6.6	60.0	17.9	88.0	7.7
6.6	60.0	17.6	87.0	5.5
7.4	59.0	18.3	82.0	6.1
6.6	64.0	19.7	67.0	12.6
7.3	59.0	19.0	68.0	15.1

# pressure_msl	# precipitation	# hour	# day_of_week	# month
1015.9	0.0	0	0	9
1015.6	0.0	1	0	9
1016.0	0.0	2	0	9
1016.3	0.0	3	0	9
1016.2	0.0	4	0	9

# quarter	# pm2_5_lag_1h	# pm2_5_lag_2h	# pm2_5_lag_3h	# pm2_5_lag_6h
3	17.1	17.3	18.3	18.9
3	15.4	17.1	17.3	18.9
3	13.8	15.4	17.1	19.0
3	12.7	13.8	15.4	18.3
3	8.8	12.7	13.8	17.3

# pm2_5_lag_6h	# pm2_5_lag_12h	# pm2_5_lag_24h	# wind_dir_sin	# wind_dir_cos
18.9	11.1	18.2	-0.7986355100472928	-0.6018150231520483
18.9	12.5	15.4	0.5299192642332049	-0.848048096156426
19.0	13.4	14.1	0.7071067811865476	-0.7071067811865475
18.3	15.5	12.0	0.8910065241883679	-0.45399049973954675
17.3	17.0	10.9	0.8829475928589269	-0.4694715627858909

8.3 Data Sources

The air quality data for this project will be sourced from the **OpenWeatherMap Air Pollution API**, which provides open access to real-time and historical global air pollution measurements. The API offers pollutant concentrations (such as PM2.5, PM10, NO2, SO2, O3, CO) along with location and timestamp information for monitoring stations in **Sofia**. Data is retrieved programmatically using **Python**.

Meteorological data for Sofia will be obtained from public weather APIs to supplement air quality measurements and improve predictive modeling.

This combination of **OpenWeatherMap Air Pollution API** and **Open-Meteo Weather API** provides a comprehensive and reliable dataset for air pollution prediction.

8.4 Data Diversity

Data will cover different locations around Sofia, all seasons, and times of day, capturing a wide range of pollution levels and weather conditions to ensure model generalizability.

8.5 Ethical Considerations

The project uses publicly accessible aggregated data, respecting privacy and confidentiality. Predictions aim to empower citizens with better information to protect health, particularly vulnerable groups like children, the elderly, and those with respiratory diseases. The model will not collect personal user data, minimizing ethical risks and ensuring compliance with data protection standards.